

Cubic-Rate Irrationality and Reciprocal-Tail Rigidity

Erdős Problem #243

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ABSTRACT

A rational reciprocal sum with $a_{n+1} \sim a_n^2$ gives an exact sequence of positive integer tail numerators. We use this integrality to prove irrationality when $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$: the numerators would have to be a cubic polynomial, but the tail recurrence forces a square condition in a cubic number field that leaves two profiles, both impossible modulo seven. We give this argument, including its use of Chebotarev, before treating the separate bounded-increment criterion for an eventual Sylvester recurrence. A weighted first-crossing criterion then identifies an estimate still needed in the unrestricted Erdős problem. None of these additional rate or increment hypotheses is deduced from quadratic growth and rationality alone.

1 The problem and the cubic rate

The identity

$$\frac{1}{a-1} = \frac{1}{a} + \frac{1}{a^2-a}$$

explains why the Sylvester recurrence $a_{n+1} = a_n^2 - a_n + 1$ produces a rational reciprocal tail. We call a sequence satisfying this recurrence eventually a *Sylvester tail*; at every sufficiently late index its tail sum is $1/(a_n - 1)$. The question is whether quadratic growth allows any other rational reciprocal sum.

Problem 1.1 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

This is the question recorded by Erdős and Graham [5, p. 64], by Erdős [6, p. 105], and as #243 in Bloom's catalogue [11]. It is not resolved here. Our leading result rules out rationality under a specified higher-order rate; the unrestricted problem is not an instance of that rate theorem.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Theorem 1.2 (cubic-rate irrationality). *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

The hypothesis still gives $a_{n+1} \sim a_n^2$, but it excludes a Sylvester tail: the latter has $a_n^2/a_{n+1} - 1 = O(1/a_n)$, much smaller than $3/n$. The theorem therefore settles the rationality question in this restricted class by proving irrationality, not by deriving a recurrence for a rational example.

An example beyond bounded increments. Set $a_1 = 8$ and

$$a_{n+1} = \left\lceil \frac{na_n^2}{n+3} \right\rceil.$$

The first terms are 8, 16, 103, 5305. Induction gives $a_n \geq 4 \cdot 2^{2^{n-1}}$, since $a_{n+1} \geq a_n^2/4$. The rounding error is less than one, so

$$0 \leq 1 + \frac{3}{n} - \frac{a_n^2}{a_{n+1}} < \frac{16}{a_n^2} = o(n^{-3}).$$

Thus its reciprocal sum is irrational by Theorem 1.2. If $P_n = \prod_{j < n} a_j$ and $t_n = P_n/a_n$, then $t_{n+1}/t_n = a_n^2/a_{n+1}$. Comparison with $n(n+1)(n+2)$, using the summable rounding errors, gives $t_n \sim Kn^3$ for some $K > 0$ and $t_{n+1} - t_n \sim 3Kn^2$. This example lies outside the bounded-increment criterion proved in Section 4.

Why a cubic is restrictive. Clearing the rational tails gives positive integers C_n with $C_{n+1}/C_n = 1 + 3/n + o(n^{-3})$. The exact recurrence, not informal differentiation of an error term, makes their fourth differences vanish eventually. Hence $C_n = An(n+1)(n+2) + B$. Arithmetic then enters: a stable gcd gives a primitive recurrence, which forces a square modulo almost every prime at every root of the cubic. Chebotarev lifts this condition to its cubic number field. A trace calculation leaves two normalised cubics; a four-term calculation modulo seven excludes both. Sections 2–3 give these steps.

The other results concern rational tails under weaker growth information. An upper bound on the increments of P_n/a_n is sufficient for a Sylvester tail (Corollary 4.6); a weighted sum over new maxima of an LCM numerator gives an exact criterion (Theorem 6.1). Their hard step is a Chinese-remainder first-crossing argument, not the number-field argument for the cubic. The remainder construction is compared with Koizumi below, and the bounded-increment proof with Bado at its point of use. The scalar summability argument is elementary and has an existing working-report antecedent; we do not claim it as a new method.

Evidence and reading route. The proofs in this paper are ordinary mathematical arguments. The margin links identify recorded Lean declarations and Comparator checks, not new executions or an independent review of this revision. In particular, the printed cubic proof uses Chebotarev; the recorded Lean route uses a Dedekind-zeta pole comparison. The [companion record](#), Section 2, gives the full cubic and more general

polynomial-extraction arguments, and its final result map locates the other extensions. The frozen evidence record retains source-pin and index gaps; they are not assertions that the corresponding theorems are false.

2 Integer tails and the information they retain

Assume for this section that a_n are strictly increasing positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n = p/q$ with positive integers p, q . Write

$$x_n = \sum_{k \geq n} \frac{1}{a_k}, \quad P_n = \prod_{j < n} a_j, \quad D_n = qP_n, \quad C_n = D_n x_n.$$

Then

$$C_n = pP_n - q \sum_{k < n} \frac{P_n}{a_k} \in \mathbb{N}_{>0}, \quad C_{n+1} = a_n C_n - D_n, \quad D_{n+1} = a_n D_n.$$

These are exact denominator-clearing identities; the fraction is not reduced at each step. They are Koizumi's integer-tail coordinates up to a common rescaling and the stated indexing [10, Lemma 4, pp. 11–12].

For large n , $a_{n+1} \geq a_n^2/2 \geq 2a_n$. The terms after $1/a_{n+1}$ sum to at most $2/a_{n+2} \leq 4/a_{n+1}^2$, so

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}), \quad \frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n). \quad (1)$$

The growth is at least double exponential after a fixed initial index. In particular, $1/a_n = o(n^{-k})$ for every fixed k . A finite prefix can be removed when proving an eventual recurrence, but the original one-based index is retained in the cubic rate: shifting n changes its lower-order terms.

2.1 The error and its zero state

Define

$$E_n = D_n - (a_n - 1)C_n.$$

An *exact state* is a collection satisfying this definition and the two displayed updates. Thus $E_n = 0$ means precisely $x_n = 1/(a_n - 1)$ at a positive tail. We use $z_+ = \max(z, 0)$, and *strict centring* means $|E_n| < C_n$.

Proposition 2.1 (error identities). *For an exact integer state,*

$$C_{n+1} = C_n - E_n, \quad (a_{n+1} - a_n^2 + a_n - 1)C_{n+1} = a_n^2 E_n - E_{n+1}.$$

Proof. Substitute the definition of E_n into the two exact updates and eliminate D_n . No limit or coprimality is used. $\square$

Equation (1) now gives $E_n/C_n = 1 - C_{n+1}/C_n \rightarrow 0$, hence strict centring eventually. Koizumi also obtains the relative-gap limit on an eventual pseudo-greedy tail [10, Corollary 3, p. 9]; his rounding convention gives the stronger half-width centring, which we do not need.

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Theorem 2.2 (absorption and descent). *For a positive exact state with strict centring, $E_n = 0$ implies $E_{n+1} = 0$. For any positive integer state with $C_{n+1} = C_n - E_n$, eventual nonnegativity of E_n implies its eventual vanishing.*

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Proof. If $E_n = 0$, the second identity in Proposition 2.1 makes E_{n+1} a multiple of C_{n+1} . Strict centring at that successor index forces the multiple to be zero. For the second assertion, C_n is eventually a nonincreasing sequence of positive integers and stabilises. $\square$

Corollary 2.3 (two zero errors). *If $E_n = E_{n+1} = 0$ and $C_{n+1} \neq 0$, then $a_{n+1} = a_n^2 - a_n + 1$. Thus eventual zero error in a positive exact state implies the eventual Sylvester recurrence.*

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Proof. Cancel the nonzero factor C_{n+1} in the second error identity. $\square$

Consequently, beyond a centring threshold either all later errors vanish, or none does. In the latter case $|E_n| \geq 1$, so the relative-error limit forces $C_n \rightarrow \infty$. This is stronger than a scalar approximation: $C_n = n + 1$, $E_n = -1$ obey the numerator update and relative limit but need not satisfy any denominator recurrence.

2.2 What reduction preserves

The gcds $G_n = \gcd(C_n, D_n)$ form a divisibility chain, since both updates preserve common divisors. Also $G_n \mid E_n$. If G_n is constant, say g , on a tail, division by g gives

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n, \quad \gcd(u_n, v_n) = 1.$$

Here $u_n > 0$ and $v_n \geq 0$; we call this a *reduced exact tail*.

Proposition 2.4 (persistent coprimality). *In a reduced exact tail, $\gcd(a_n, v_n) = 1$. Distinct multipliers are pairwise coprime, and every earlier multiplier is coprime to every later numerator.*

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Proof. A prime dividing both a_n and v_n would divide u_{n+1} and v_{n+1} , contrary to reducedness. For $i < t$, the factor a_i divides v_t . Reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. $\square$

The cubic proof will make G_n bounded by a finite difference; the bounded-increment proof will bound it using negative errors. These are different reasons for reaching the same primitive recurrence.

3 Proof of cubic-rate irrationality

Suppose, towards a contradiction, that the reciprocal sum in Theorem 1.2 is rational. Construct C_n, D_n as in Section 2. Equation (1) transfers the rate without changing its precision:

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}). \quad (2)$$

We first extract the polynomial, then use the exact denominator dynamics to exclude it. The following is the cubic specialisation of the companion's Section 2; its general extraction result is not needed here.

3.1 An eventual polynomial from an integer sequence

Put $F_n = n(n+1)(n+2)$, so $F_{n+1}/F_n = 1 + 3/n$, and write the error in (2) as $\varepsilon_n = o(n^{-3})$. For $z_n = C_n/F_n$,

$$\frac{z_{n+1}}{z_n} = 1 + \frac{\varepsilon_n}{1 + 3/n}.$$

The logarithms of these positive ratios are absolutely summable. Thus $z_n \rightarrow A > 0$; their tails are $o(n^{-2})$, so $z_n - A = o(n^{-2})$. Hence $\delta_n = C_n - AF_n = o(n)$. It would be invalid to differentiate this last estimate formally. Instead the recurrence gives

$$\Delta\delta_n = \frac{3}{n}\delta_n + \varepsilon_n C_n = o(1), \quad \Delta h_n = h_{n+1} - h_n.$$

It follows that $\Delta^4 C_n = \Delta^3(\Delta\delta_n) \rightarrow 0$. These fourth differences are integers and therefore vanish eventually. The sequence C_n agrees on a tail with a polynomial in $\mathbb{Q}[n]$ of degree at most three. Since $C_n - AF_n = o(n)$, comparison of polynomial coefficients gives

$$C_n = An(n+1)(n+2) + B, \quad A \in \mathbb{Q}_{>0}, \quad B \in \mathbb{Q}. \quad (3)$$

The little-oh hypothesis has done real work. The positive integer sequence $F_n + (-1)^n$ has ratio $1 + 3/n + O(n^{-3})$ but is not eventually a polynomial. This scalar example explains the precision in the extraction step; it is not a reciprocal-tail counterexample.

3.2 Normalisation and irreducibility

For a sufficiently late n , the integer $G_n = \gcd(C_n, D_n)$ divides $C_n, C_{n+1}, C_{n+2}, C_{n+3}$ and therefore divides $\Delta^3 C_n = 6A$. The positive divisibility chain G_n is bounded and stabilises at g . Proposition 2.4 applies to $u_n = C_n/g, v_n = D_n/g$. In addition, adjacent numerators are coprime: $\gcd(u_n, u_{n+1}) = \gcd(u_n, v_n) = 1$.

Write the polynomial for u_n as

$$Q(n) = \frac{m}{6}n(n+1)(n+2) + c. \quad (4)$$

Its third difference $m = 6A/g$ is a positive integer. Since Q is integer-valued at all sufficiently large integers, it is integer-valued at every integer: choose a common denominator of its coefficients and translate any integer by a sufficiently large multiple of that denominator. In particular $c = Q(0) \in \mathbb{Z}$. If a prime ℓ divides c , choose an arbitrarily late $n \equiv -1 \pmod{6\ell}$. Both $Q(n)$ and $Q(n+1)$ would be divisible by ℓ , contradicting adjacent coprimality. This also excludes $c = 0$. Thus

$$m \in \mathbb{N}_{>0}, \quad c \in \{1, -1\}.$$

The multipliers $a_n > 1$ on the reduced tail are pairwise coprime, so infinitely many distinct primes occur among them. Once a prime divides one such a_n , it divides every later v_t and no later u_t . If the cubic Q had a rational root, that root would reduce to a root modulo all but finitely many primes. Choose one of the persistent primes outside those exceptions; a sufficiently late index in the root's residue class would have $\ell \mid Q(t) = u_t$, a contradiction. A cubic is reducible over $\mathbb{Q}$ only if it has a rational root. Hence Q is irreducible.

Set $\kappa = m/6$, $\eta = 6c/m$, and

$$f(T) = T^3 - T + \eta, \quad Q(n) = \kappa f(n+1).$$

Let α be a root of f and $K = \mathbb{Q}(\alpha)$, a cubic number field.

3.3 A modular square and its number-field consequence

Eliminating v_n from the reduced recurrence gives

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \quad (5)$$

Take a prime $\ell \nmid 6m$ and a root r of f in $\mathbb{F}_\ell$. The roots $0, 1, -1$ are excluded because $\eta \neq 0$ modulo ℓ . Choose a late index k with $k+1 \equiv r \pmod{\ell}$. Then $u_k \equiv 0$ and

$$f(r-1) = -3r(r-1), \quad f(r+1) = 3r(r+1).$$

Multiplying (5) at index $k-1$ by u_{k-1} yields

$$-u_{k-1}u_{k+1} = (a_{k-1}u_{k-1})^2 = 9\kappa^2 r^2(r^2 - 1) \quad \text{in } \mathbb{F}_\ell.$$

The expression is nonzero. Therefore $r^2 - 1$ is a nonzero square for every root r of f modulo every such prime.

We spell out the step from primes to K . Suppose $\alpha^2 - 1$ were not a square in K , and let $\beta^2 = \alpha^2 - 1$. The nontrivial automorphism of the quadratic extension $K(\beta)/K$ fixes α and exchanges β and $-\beta$. Extend it to an automorphism σ of a Galois closure over $\mathbb{Q}$. By the Chebotarev density theorem [9, §3, author-version p. 15], infinitely many unramified rational primes have Frobenius in the conjugacy class of σ . Choose a prime above each of them whose Frobenius is σ , and discard the finitely many primes of bad reduction, residue characteristic two, or vanishing denominators.

In the residue field, Frobenius fixes $\bar{\alpha}$ but sends $\bar{\beta}$ to $-\bar{\beta} \neq \bar{\beta}$. Thus $\bar{\alpha} \in \mathbb{F}_\ell$ is a root of f , while $\bar{\alpha}^2 - 1$ is not a square in $\mathbb{F}_\ell$: its only two square roots in the residue field are $\pm\bar{\beta}$, neither fixed by Frobenius. This contradicts the modular condition. Consequently there is a $\beta \in K$ with $\beta^2 = \alpha^2 - 1$. This is the global ingredient; the remaining calculation is algebraic.

3.4 The trace calculation leaves only two cubics

Put $z = \alpha + \beta$. Then $z^{-1} = \alpha - \beta$ and $\alpha = (z + z^{-1})/2$, so $\mathbb{Q}(z) = K$. Write the minimal polynomial of z as $z^3 + bz^2 + dz + w$, with $b, d, w \in \mathbb{Q}$ and $w \neq 0$. The polynomial for α gives

$$\text{Tr}(\alpha) = 0, \quad \text{Tr}(\alpha^2) = 2, \quad \text{Tr}(\alpha^3) = -3\eta.$$

Newton's identities for z and z^{-1} turn these three equations into

$$d = -bw, \quad b^2 + b(w - w^{-1}) = 1, \quad \eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \quad (6)$$

For clarity, the first uses $\text{Tr}(z) = -b$, $\text{Tr}(z^{-1}) = -d/w$. After substituting $d = -bw$, the second follows from $4\text{Tr}(\alpha^2) = \text{Tr}(z^2) + 6 + \text{Tr}(z^{-2})$; the third follows from $8\text{Tr}(\alpha^3) = \text{Tr}(z^3) + \text{Tr}(z^{-3})$, since $\text{Tr}(z + z^{-1}) = 0$.

The middle equation factors as $(bw - 1)(b + w) = 0$. Write $w = r/s$ in lowest terms, with $r \neq 0, s > 0$. Substituting $b = -w$ or $b = 1/w$ into the last equation and using $\eta = 6c/m$ gives respectively

$$m = \frac{48c rs^3}{(r^2 + s^2)^2} \quad \text{or} \quad m = \frac{48c r^3 s}{(r^2 + s^2)^2}.$$

Because $\gcd(r^2 + s^2, rs) = 1$ and m is an integer, $(r^2 + s^2)^2 \mid 48$. The only possible nonzero coprime pair has $r^2 + s^2 = 2$, hence $|r| = s = 1$; positivity then gives $m = 12$. The only remaining numerators are

$$Q(n) = 2n(n+1)(n+2) + 1 \quad \text{and} \quad Q(n) = 2n(n+1)(n+2) - 1.$$

3.5 Both profiles fail modulo seven

In the plus case, choose a late block starting at $n \equiv 0 \pmod{7}$; in the minus case, start at $n \equiv 1 \pmod{7}$. The four successive numerators have residues

$$(u_n, u_{n+1}, u_{n+2}, u_{n+3}) \equiv \begin{cases} (1, 6, 0, 2) & (c = 1), \\ (4, 5, 0, 1) & (c = -1). \end{cases}$$

Call the first two denominator residues d_0, d_1 and the displayed numerator residues $c_0, c_1, 0, c_3$. From $0 = a_{n+1}c_1 - d_1$ and $c_3 = -a_{n+1}d_1$ we get $d_1^2 = -c_1c_3 = 2$. Thus $d_1 \in \{3, 4\}$. The first update, however, gives

$$d_1 = a_n d_0 = \frac{d_0(d_0 + c_1)}{c_0}.$$

As d_0 runs through $\mathbb{F}_7$, the plus case is $d_0(d_0 + 6)$ and the minus case is $2d_0(d_0 + 5)$. Both have image $\{0, 2, 5, 6\}$, disjoint from $\{3, 4\}$. This contradiction excludes both profiles and proves Theorem 1.2.

The companion proves the stronger exclusion of agreement with any such cubic outside a set of density zero. Its positive disagreement bound may depend on the polynomial; no universal numerical lower bound is asserted here. For this theorem, exclusion of eventual agreement suffices.

4 A separate route: bounded upward increments

The cubic argument exploits a very precise rate. We now return to the weaker assumption $a_{n+1} \sim a_n^2$ and ask what extra control forces an eventual Sylvester recurrence. The useful condition is one-sided: large falls of the integer numerator are allowed, but its rises are bounded. This section does not use the number-field argument.

Theorem 4.1 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ and $E : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy*

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1. $a_n > 1$ and $C_n > 0$ for every n ;
2. the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;
3. $E_n = D_n - (a_n - 1)C_n$ for every n ;
4. eventual strict centring: $|E_n| < C_n$ for all large n ;
5. eventually bounded negative part: $-B \leq E_n$ for all large n , for some integer $B \geq 0$;
6. vanishing relative error: for every integer $K \geq 1$ there is an N with $K|E_n| < C_n$ for all $n \geq N$.

Then $E_n = 0$ for all sufficiently large n .

Condition (6) is $|E_n|/C_n \rightarrow 0$ and already implies (4). The redundant centring condition identifies exactly what absorption uses. For rational tails Section 2 supplies every condition except (5), after restriction to a sufficiently late tail. The error bound is an extra arithmetic hypothesis, not a consequence of relative convergence.

4.1 A stable gcd and a forbidden block

Proposition 4.2 (gcd stabilisation). *For a positive exact state, suppose that some fixed integer $B \geq 1$ satisfies $-B \leq E_n < 0$ at infinitely many indices. Then $G_n = \gcd(C_n, D_n)$ is eventually constant. Division by its stable value gives a reduced exact tail.*

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Proof. We have $G_n \mid G_{n+1}$ and $G_n \mid E_n$. Given n , choose $t \geq n$ with $-B \leq E_t < 0$. Then $G_n \leq G_t \leq -E_t \leq B$. The bounded positive divisibility chain is eventually constant; dividing by its stable value preserves both updates and gives coprime states. $\square$

A bound at arbitrarily late negative indices suffices for this gcd argument. To control every upward step we will still need the lower bound on every sufficiently late error in Theorem 4.1.

Lemma 4.3 (consecutive multiples). *For pairwise coprime integers $m_0, \dots, m_{B-1} \geq 2$ and every lower bound, there is a larger t such that $m_i \mid t + i$ for each $i < B$.*

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Proof. Solve $t \equiv -i \pmod{m_i}$ by the Chinese remainder theorem and add multiples of $\prod_i m_i$ to exceed the prescribed bound. $\square$

For instance, integers coprime to both 2 and 3 cannot occupy either $6k + 2$ or $6k + 3$. Upward jumps of size at most two cannot cross that pair from below. The next statement isolates this elementary mechanism; it does not require the path to be monotone.

Theorem 4.4 (Chinese remainder theorem and first crossing). *Let $u : \mathbb{N} \rightarrow \mathbb{N}$ tend to infinity and let $B \geq 1$ be an integer with $u_{n+1} \leq u_n + B$ for every n . There is no sequence of pairwise coprime integers $m_i \geq 2$ for which $\gcd(m_i, u_i) = 1$ whenever $i < t$.*

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Proof. Apply Lemma 4.3 to $m_0, \dots, m_{B-1}$, choosing $t > \max(u_0, \dots, u_B)$. There is a first $n > B$ with $u_n \geq t$. Then

$$t \leq u_n \leq u_{n-1} + B < t + B.$$

Thus $u_n = t + i$ for some $i < B < n$, and $m_i \mid u_n$, contrary to $\gcd(m_i, u_i) = 1$. $\square$

Bado uses the same forbidden-block and first-crossing mechanism under a two-sided bounded-error hypothesis [7, Theorem 5.1, pp. 4–5]. Here the separate gcd argument allows the lower error bound alone. This comparison identifies the change in hypotheses without treating the Chinese remainder theorem itself as a new method.

Proof of Theorem 4.1. Suppose E_n does not vanish eventually. Absorption excludes all sufficiently late zeros, so $|E_n| \geq 1$; condition (6) gives $C_n \rightarrow \infty$. There must be infinitely many negative errors, since otherwise integer descent would give eventual zero. Their magnitudes are bounded by B at late indices, so $B \geq 1$ and Proposition 4.2 gives a stable gcd g . On the reduced tail, $u_n = C_n/g \rightarrow \infty$ and

$$u_{n+1} - u_n = -E_n/g \leq B.$$

By Proposition 2.4, the multipliers $a_i \geq 2$ are pairwise coprime and each is coprime to every later numerator. They supply the moduli prohibited by Theorem 4.4, a contradiction. $\square$

Corollary 4.5. *Under Theorem 4.1, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ eventually.*

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Proof. Theorem 4.1 gives eventual zero error; positivity of C_n and Corollary 2.3 give the recurrence. $\square$

4.2 The condition in the original sequence

Let $\gamma_n = a_n^2/a_{n+1} - 1$. For $t_n = P_n/a_n$,

$$t_{n+1} - t_n = t_n \gamma_n.$$

Thus the following condition bounds increments, not the size of t_n . It imposes no lower bound on those increments and no monotonicity.

Corollary 4.6 (bounded increments of P_n/a_n). *Let $a_1 < a_2 < \dots$ be positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j < n} a_j$. If*

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$$\limsup_{n \rightarrow \infty} \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < +\infty,$$

then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .

Proof. For the canonical integer tail, direct substitution gives

$$E_n + q \frac{P_n}{a_n} \gamma_n = q P_n \left(\frac{1}{a_{n+1}} - (a_n - 1) \sum_{k \geq n+2} \frac{1}{a_k} \right). \quad (7)$$

By the tail estimate, the subtracted term is at most $4(a_n - 1)/a_{n+1}^2 < 1/a_{n+1}$ eventually. The right side is positive, so an upper bound on $(P_n/a_n)\gamma_n$ gives a lower bound on E_n . All the remaining hypotheses were established in Section 2. Apply Theorem 4.1 and Corollary 4.5. $\square$

Koizumi's proof of Corollary 4(1) uses this product-weighted comparison [10, (11)–(12), p. 15]; that corollary assumes a nonpositive upper limit, whereas the argument here allows any finite upper limit. A corresponding LCM comparison appears below. For $a_n = b^{2^{n-1}}$, $b \geq 2$, the increments of P_n/a_n are zero. Since this is not a Sylvester tail, the corollary gives irrationality. For a Sylvester sequence starting at $a_1 > 1$, $a_n - 1 = (a_1 - 1)P_n$ and the increments tend to zero.

The inclusive rate $1/n$ gives another useful sufficient condition.

Corollary 4.7 (an inclusive one-sided $1/n$ bound). *Let $a_1 < a_2 < \dots$ be positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum_n 1/a_n \in \mathbb{Q}$. Suppose that for some $K \geq 0$ and $\varepsilon > 0$,*

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$$\gamma_n := a_n^2/a_{n+1} - 1 \leq \frac{1}{n} + \frac{K}{n^{1+\varepsilon}} \quad \text{for all large } n.$$

Then the sequence is eventually Sylvester. In particular, the conclusion holds under the pointwise eventual bound $\gamma_n \leq 1/n$.

Proof. The hypothesis bounds the positive part of γ_n eventually by $1/n + K/n^{1+\varepsilon}$. Since $t_{n+1}/t_n = 1 + \gamma_n$, comparison with n and the convergent product of $1 + K/n^{1+\varepsilon}$ gives $t_n = O(n)$. Hence $t_n \gamma_n \leq t_n(\gamma_n)_+ = O(1)$, and Corollary 4.6 applies. $\square$

The summable remainder is important: merely $\limsup n(\gamma_n)_+ \leq 1$ need not give this product estimate. For the scalar ratios $1 + \gamma_n = 1 + (1 + 1/\log n)/n$, the product grows on the scale $n \log n$, not n . This is a limitation of that inference, not an arithmetic counterexample.

For a sequence with genuinely positive limiting increments, take $a_1 = 4$ and $a_{n+1} = \lceil na_n^2/(n+1) \rceil$, beginning 4, 8, 43, 1387. Here $a_n \geq 2 \cdot 2^{2^{n-1}}$ and

$$0 \leq 1 + \frac{1}{n} - \frac{a_n^2}{a_{n+1}} < \frac{4}{a_n^2}.$$

Consequently $t_n \sim Kn$ and $t_{n+1} - t_n \rightarrow K > 0$. The corollary proves irrationality because $\gamma_n \sim 1/n$ is incompatible with a Sylvester tail. The cubic example in the introduction has unbounded increments and requires the different proof in Section 3.

5 Finite total relative increase

The next condition concerns the sum of all upward increments of C_n , each divided by the value before that step. A Sylvester tail has no such increments after a finite index. In contrast, $C_n = n + 1$ and $C_n = (n + 1)^2$ both have divergent sums, although their relative increments tend to zero. Summability is thus much stronger than that limit. The denominator recurrence is not needed for the following argument.

Theorem 5.1 (a convergent sum of relative increases). *Let C_n be positive integers and E_n integers satisfying $C_{n+1} = C_n - E_n$. If*

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$$\sum_n \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. Neither denominator dynamics nor vanishing relative error is required.

Proof. Set $\delta_n = (-E_n)_+ / C_n$. Then

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer upper bound K for C_n . Each strict rise contributes at least $1/K$ to $\sum \delta_n$, so there are only finitely many rises. The remaining positive integer sequence is nonincreasing and stabilises, and the update then gives $E_n = 0$. $\square$

The proof also allows positive real C_n with integer E_n : the values lie in $C_0 + \mathbb{Z}$, whose bounded positive part is finite. Discrete increments are essential to this argument. With real errors, $C_n = 1 + 1/(n+1)$ decreases forever with zero relative-increase sum; without positivity, $C_n = -n - 1$, $E_n = 1$ does so too. The companion's Section 13 gives the details.

On an exact reciprocal-tail orbit the conclusion gives the Sylvester recurrence. For the gap sequence of the pseudo-greedy expansion, the same criterion, with the same product bound and integer descent, appears in the Erdős Problem a Day working report on Problem #243, dated 12 August 2026 [8, "A global termination criterion"]; the statement above is for any positive integer sequence with $C_{n+1} = C_n - E_n$. This summability condition and the bound in Theorem 4.1 are different estimates. For exact reciprocal tails with $|E_n|/C_n \rightarrow 0$, each is equivalent to eventual vanishing of E_n ; neither estimate has been derived here from the unrestricted problem.

6 New maxima of an LCM numerator

Instead of multiplying all earlier denominators, one can clear the rational tail with their least common multiple. We will sum only at steps where the resulting numerator exceeds all its earlier values. This allows some increases to be omitted from the sum and others to receive smaller weights. Set

$$L_n = \text{lcm}(q, a_1, \dots, a_{n-1}), \quad M_n = D_n / L_n, \quad U_n = C_n / M_n, \quad V_n = E_n / M_n.$$

The rational tail has denominator dividing L_n , so U_n and V_n are integers. This need not be a reduced fraction: U_n and L_n may share a factor. With $\rho_n = \gcd(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

These coordinates and updates also appear in Bado's September 2026 preprint. With his denominator parameter equal to q , his M_{n-1} , Δ_{n-1} , K_{n-1} , u_n and g_n are respectively our L_n , M_n , U_n , V_n and ρ_n ; his (19) is the middle update above [7, Proposition 7.1 and (16)–(20), pp. 6–7].

Since $L_n \mid qP_n$ and $V_n = (L_n / (qP_n))E_n$, multiplying (7) by $L_n / (qP_n)$ gives, eventually,

$$0 < V_n + \frac{L_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) \leq \frac{L_n}{a_{n+1}}. \quad (8)$$

Its lower bound gives the LCM corollary below. Summability of the comparison error is proved when needed for the weighted criterion; the different denominator update is accounted for by ρ_n .

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Strict centring already gives $U_{n+1} < U_n$ when $\rho_n \geq 2$, so every sufficiently late strict rise has $\rho_n = 1$. The stronger eventual bound $-U_n \leq 2V_n$, supplied by $V_n/U_n = E_n/C_n \rightarrow 0$, gives the quantitative estimate $U_{n+1} \leq 3U_n/4$ when $\rho_n \geq 2$. At a sufficiently late record step, where $\rho_n = 1$, the actual jump is $d_n = U_{n+1} - U_n = -V_n > 0$. This identity is not asserted at a contracting step with $\rho_n \geq 2$.

Theorem 6.1 (a convergent weighted sum over new maxima). *Assume the growth and rationality hypotheses of Problem 1.1. Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

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$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Only a step reaching a new maximum contributes. At a late such step, $-V_n = U_{n+1} - U_n$, so the first B units of the increase are ignored. A Sylvester tail makes the sum finite because $V_n = 0$ eventually. On a non-Sylvester tail, the proof gives divergence for every admissible f and every fixed B . The limit $V_n/U_n \rightarrow 0$ gives a pointwise relative-error estimate; no bound on this weighted sum is derived from that limit here. The restriction on f matters: $f(t) = 1/t$ is permitted, as is $1/[t \log(et)]$, but $1/t^2$ is not. For the latter weight, the integral in the crossing argument is finite and cannot contradict unbounded growth.

Proof. Suppose the sequence is not eventually Sylvester. Absorption and integrality give

$$\frac{1}{U_n} \leq \frac{|V_n|}{U_n} = \frac{|E_n|}{C_n} \rightarrow 0.$$

Thus $U_n \rightarrow \infty$, so there are infinitely many record steps. At every sufficiently late one, $\rho_n = 1$, so a_n is coprime to L_n . The corresponding terms a_n are pairwise coprime: an earlier term divides the later L_n , while the term at that record is coprime to L_n . For any fixed $B \geq 1$, choose B such multipliers $m_0, \dots, m_{B-1} > B$ and take T after their indices and after the threshold beyond which records have $\rho_n = 1$. Their size follows from $a_n \rightarrow \infty$.

Put $P = \prod_i m_i$ and choose x by the Chinese remainder theorem with $m_i \mid x + i$. Consider all translates $\tau = x + B + kP > R_T$, $k \in \mathbb{Z}$; the first is at most $R_T + P$. We first show that crossing one of these heights requires a jump larger than B , then count how many heights a single jump can cross. Here $d_n = U_{n+1} - U_n$ is the actual jump, including any part that recovers an earlier decrease. A first crossing $U_n \leq R_n < \tau \leq U_n + d_n$ is a record step. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so some m_i divides U_n . It also divides L_n , hence divides $d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

If the step first crosses $h \geq 1$ such heights, their spacing gives $(h - 1)P < d_n$. With $r = d_n - B \geq 1$ and $P \geq B + 1$, we have $d_n = B + r \leq Pr$, whence $h \leq r$. Monotonicity of f now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each selected height above R_T has exactly one first crossing. The first selected height is at most $R_T + P$; each later height is P further on. For $R_N \geq R_T + P$, comparison of each interval with its left endpoint gives

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T+P}^{R_N} f(t) dt. \quad (9)$$

Since $R_n \rightarrow \infty$, the right side diverges for every $B \geq 1$; $B = 0$ follows by domination. Conversely a Sylvester tail telescopes to $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (9) is $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n$. Here summability, not merely a one-sided bound, is needed. Since $a_n x_n \rightarrow 1$, we have $C_n/a_n \sim qP_n/a_n^2$, and

$$\frac{P_{n+1}/a_{n+1}^2}{P_n/a_n^2} = \frac{a_n^3}{a_{n+1}^2} \rightarrow 0.$$

The ratio test gives $\sum_n C_n/a_n < \infty$; now $U_n \leq C_n$ and $f(U_n) \leq f(1)$ make the comparison error summable. The errors in (7) and (8) are also absolutely summable, since each is eventually between zero and $qP_n/a_{n+1} = O(P_n/a_n^2)$. Consequently Theorem 6.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (10)$$

for some B . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. Also, d_n is the actual jump, including any recovery from a drawdown; it must not be replaced by $R_{n+1} - R_n$ in the crossing proof.

Corollary 6.2 (a bound using the least common multiple). *Assume the hypotheses of Problem 1.1. Write $A_n = \text{lcm}(a_1, \dots, a_{n-1})$ with $A_1 = 1$. If*

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$$\limsup_{n \rightarrow \infty} \frac{A_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < \infty,$$

then the sequence is eventually Sylvester.

Proof. Since $L_n/A_n = q/\gcd(q, A_n)$ lies in $[1, q]$, an upper bound on the expression in the corollary gives an upper bound on $(L_n/a_n)(a_n^2/a_{n+1} - 1)$; negative terms remain negative. Equation (8) therefore bounds the negative part of V_n . Choose an integer B above that eventual bound. The record series then vanishes after a finite prefix, so Theorem 6.1 applies. $\square$

Erdős and Straus assume a nonpositive upper limit in this LCM expression [1, Theorem 3, p. 132]: their N_k is A_{k+1} and their growth ratio has index $k+1$. Tijdeman and Yuan extend this type of criterion to positive numerators [4]. Here any finite upper bound suffices under the quadratic-limit assumption. Since $A_n \mid P_n$, the LCM hypothesis is no stronger than the product hypothesis in Corollary 4.6: if the product expression is at most B , the LCM expression is at most $\max(B, 0)$. When the earlier terms are pairwise coprime the two weights agree; repeated prime factors can make the LCM much smaller. This comparison does not assert the existence of a non-Sylvester rational example satisfying one bound but not the other.

The finite upper limit means an eventual upper bound; it does not require q to divide the LCM of the preceding denominators. For positive integer summand numerators b_n , the modified error is $V_n = b_n L_n - (a_n - 1)U_n$. The companion, Section 6 under “Integer coefficients”, separately proves boundedness from a lower bound on V_n and eventual constancy when $V_n/U_n \rightarrow 0$ is also assumed. The latter conclusion must not be read into the first hypothesis alone. The same section links finite examples separating them; compare Badea [3, p. 316] and Tijdeman–Yuan [4]. The companion’s Section 3 retains the signed-series comparison with Duverney [2, Corollary 3.2, p. 287] and the different growth hypotheses for irrationality exponents in [14, Theorem 1, author-version p. 2]. Those comparisons are not additional hypotheses of the crossing theorem.

7 The remaining arithmetic estimate

Proposition 7.1 (necessary profile). *The integer tail of a sequence satisfying Problem 1.1’s hypotheses but not its conclusion has $E_n \neq 0$ eventually, $|E_n|/C_n \rightarrow 0$, unbounded negative magnitudes along negative indices, and*

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$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Proof. Absorption excludes late zeros, descent excludes an eventually nonnegative error, and Theorems 4.1 and 5.1 exclude the two finiteness conditions. $\square$

For $B \geq 0$ put

$$F_B(X) = \sum_{\substack{n \in \mathcal{R} \\ U_n \leq X}} (-V_n - B)_+.$$

For a rational tail satisfying the growth hypothesis, the weighted criterion reduces the original problem to proving that, for some fixed integer $B \geq 0$,

$$\liminf_{X \rightarrow \infty} \frac{F_B(X)}{X} = 0. \quad (11)$$

If the recurrence is not eventually Sylvester, first crossings instead give $F_B(X) \geq X/P_B - O_B(1)$, with an orbit-dependent CRT modulus P_B . The sum groups each record step by its starting numerator U_n , not by the running maximum R_n or the time index. On a non-Sylvester tail, $U_n \rightarrow \infty$, so only finitely many of these steps begin below any fixed X . The following lemma proves equivalence with the existence of an admissible weight. Enumerate the record indices $n \in \mathcal{R}$ by j , and apply it with $u_j = U_n$ and $w_j = (-V_n - B)_+$. The missing assertion is that growth and rationality force (11); the first-crossing lower bound is already proved.

Lemma 7.2 (weights and linear density). *Let u_j be positive integers, let $w_j \geq 0$, and put $F(X) = \sum_{u_j \leq X} w_j$, with the sum allowed a priori to be $+\infty$. Then $\liminf_{X \rightarrow \infty} F(X)/X = 0$ if and only if there is a finite nonincreasing $f : [1, \infty) \rightarrow [0, \infty)$ with $\int_1^\infty f(t) dt = \infty$ and $\sum_j w_j f(u_j) < \infty$.*

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Proof. Suppose the lower limit vanishes. Choose cutoffs at which the ratio $F(X)/X$ is small. Since F is constant on each $[m, m+1)$, there are integers $X_1 < X_2 < \dots$ with $X_k \geq 2^k$ and $F(X_k) \leq 2^{-k} X_k$. Set

$$f(t) = \sum_{k \geq 1} \frac{\mathbf{1}_{[1, X_k]}(t)}{X_k}.$$

This function is nonincreasing and bounded by $\sum_k 2^{-k}$. Each cutoff contributes $1 - 1/X_k$ to its integral, but at most 2^{-k} to the weighted sum. Interchanging nonnegative sums therefore gives

$$\int_1^\infty f(t) dt = \sum_k (1 - 1/X_k) = \infty, \quad \sum_j w_j f(u_j) = \sum_k \frac{F(X_k)}{X_k} \leq 1.$$

Conversely, let f be such a weight and suppose $F(m) \geq cm$ for some $c > 0$ and all integers $m \geq M$. A nonincreasing nonnegative function with divergent integral is positive everywhere, so $F(X) \leq f(X)^{-1} \sum_j w_j f(u_j)$ is finite. Partial summation gives, for $N > M$,

$$\sum_{u_j \leq N} w_j f(u_j) = F(N)f(N) + \sum_{m=1}^{N-1} F(m)(f(m) - f(m+1)) \geq c(Mf(M) + \sum_{m=M+1}^N f(m)),$$

every term being nonnegative. The right side diverges as $N \rightarrow \infty$, because $\sum_{m \geq 1} f(m) \geq \int_1^\infty f(t) dt = \infty$. $\square$

7.1 Why the scalar profile is not enough

The remaining estimate must use arithmetic compatibility, not just the size of the error. For example, $C_n = n^2 + 1$, $E_n = -(2n+1)$ has vanishing relative error and the necessary scalar pattern. But at $n = 2, 3, 4$ its numerators are 5, 10, 17. The update $10 = 5a_2 - D_2$ forces $5 \mid D_2$, hence $5 \mid D_3 = a_2 D_2$; the next update would force $5 \mid 17$. It is not an exact tail. The companion's Section 14 also constructs sequences avoiding sparse fixed

moduli and distinguishes that avoidance from compatibility with all later denominators. Neither property alone supplies (11).

The profile is a necessary condition, not a construction. For a zero-indexed exact orbit with $a_n > 1$, $C_n > 0$, $D_n \geq 0$ and $E_n/C_n \rightarrow 0$, the companion's Section 4 proves that $D_n > 0$, $C_n/D_n \rightarrow 0$ and $\sum_n 1/a_n = C_0/D_0$; it also derives quadratic growth. Thus these global orbit assumptions already make the orbit the integer tail of a reciprocal series. A scalar numerical profile or a finite admissible prefix does not.

Problem 7.3 (growth of repeated denominator factors). For every rational-tail orbit satisfying the hypotheses but not the conclusion of Problem 1.1, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \geq 1$ for which

$$2^n \leq M_n^K$$

at infinitely many indices?

Section 2 gives $C_{n+1}/C_n \rightarrow 1$; averaging logarithms therefore gives $\log C_n = o(n)$. Since $1 \leq M_n \leq C_n$, every such orbit already satisfies $\log M_n/n \rightarrow 0$. A positive answer therefore holds exactly when no such orbit exists: proving it from failure of Sylvester behaviour would exclude every counterexample.

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A A factorial residue reduction for forced orbits

A constant-negative error $E_n = -m$ with $C_0 = c$ produces $C_n = c + nm$ and the shape equation

$$D_n + m = (a_n - 1)C_n. \quad (12)$$

In the case $m = c = 1$, where (12) reads $D_n + 1 = (a_n - 1)(n + 1)$, each multiplier is determined by its predecessor; we call such an orbit *forced*. At index n the numerator of the next multiplier is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#), and the divisor is $n + 2$. The orbit continues for another step exactly when this division is exact. Here survival counts exact divisions; it does not impose the positivity conditions of an infinite reciprocal-tail orbit. The [formal survival predicate](#) requires exact division at each step.

Example A.1 (the shape equation running until it fails). The forced orbit from $a = 3$ reads this way: $\text{num}(0, 3) = 6$ is divisible by 2 and gives $a_1 = 3$, while $\text{num}(1, 3) = 13$ is not divisible by 3, so the orbit stops there.

Direct iteration can produce very large intermediate integers. To decide whether the first h divisions are exact, however, it suffices to know the initial value modulo $(h+1)!$. Computing a pseudo-greedy orbit through residues modulo a shrinking product modulus is Koizumi's method [10, Remark 2 and Algorithm 1, pp. 13–14]; for the forced numerator that product is a factorial.

Theorem A.2 (factorial residue reduction). *For all h and all integers $a \equiv b \pmod{(h+1)!}$, the orbit from a survives h forced updates exactly when the orbit from b does.*

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Proof. Define $M(0, i) = 1$ and $M(h+1, i) = (i+2)M(h, i+1)$. Thus $M(h, 0) = (h+1)!$. We prove the stronger statement that, at index i , congruent inputs modulo $M(h, i)$ survive the same h updates. There is nothing to prove for $h = 0$.

For the inductive step, suppose $a \equiv b \pmod{(i+2)M(h, i+1)}$. Since $\text{num}(i, \cdot)$ is an integer polynomial, its values at a and b are congruent modulo that product. In particular, one is divisible by $i+2$ if and only if the other is. If neither is divisible, both orbits stop. Otherwise their quotients are congruent modulo $M(h, i+1)$, so the induction hypothesis applies to the remaining h updates at index $i+1$. Taking $i = 0$ proves the claim. $\square$

The formal [factorial residue reduction](#) uses the [shrinking-modulus induction](#), [polynomial congruence](#), and [cancellation after exact division](#). Its modulus is identified by the [ascending-factorial formula](#) and its [factorial value at the initial index](#).

At $h = 1$, survival means $2 \mid a^2 - 2a + 3$, or equivalently that a is odd. This illustrates Theorem A.2 with modulus $2! = 2$.

The finite enumeration recorded in the companion over $2 \leq a_0 < 5000$ gives at most 17 successful updates (18 values including a_0). The omitted seed $a_0 = 1$ is a fixed point, outside $a_n \geq 2$. This finite search is not a proof of the [constant-negative exclusion](#), which concerns infinite natural-number orbits under its stated hypotheses. Example 9.1 of the companion gives the further search details.

B Proof record and related work

The companion *Reciprocal-Tail Rigidity: Theorems, Proofs and Questions* keeps the complete arguments and fixed-revision declaration routes. Its Section 2 contains the general regular-rate extraction, cubic square specialisation, trace calculation and the stronger positive-density disagreement result. Sections 4–5 develop exact tails; Sections 11–12 give the gcd and bounded-increment arguments; Section 6 contains the LCM and integer-coefficient results; Section 13 treats the scalar summability criterion; and Section 14 keeps the unresolved arithmetic estimates and failed constructions. Its final result map records the distinct proof dependencies. Nothing in a source link turns an auxiliary numerical construction into a rational reciprocal series.

The older integer-remainder method is represented by Erdős–Straus [12, Theorem 2.1, pp. 85–86]. The polynomial Cantor-series hypotheses in Hančl–Tijdeman [13, Theorem 2.2, pp. 39–40] and the freely selectable series of Problem #270 [15, Theorems 1–2] concern different settings; the companion's Section 3 explains the distinction. The Isabelle/HOL work [16] formalises older Erdős–Straus criteria, not Problem #243. The

general regular-rate extraction in the companion also uses the standard Gamma-ratio asymptotics recorded in [17]; the cubic specialisation here uses only its explicit polynomial F_n .

The generated margin links and the authored source manifest retain the original revision identities. They separate formal propositions from this paper's ordinary proofs; recorded checks have not been rerun for this editorial revision. The number-field route printed here invokes Chebotarev, whereas the recorded formal square-specialisation route uses the simple pole of the Dedekind zeta function. Neither route gives the missing unrestricted estimate in Section 7.

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